

DoME Project Report

Team NSUA

Design of a Multipurpose Sieving Machine Power Transmission System

1. Problem Statement

Design and fabricate a mechanically operated sieving machine capable of separating granular materials into at least two size categories (coarse and fine).

The machine must operate at a uniform vibration speed, handle a load of 2–3 kg of material, and perform continuous sieving with minimal human effort.

The design should be structurally stable, easy to operate, and cost-efficient. The machine should include a suitable vibration mechanism, mesh selection arrangement, and supporting frame.

Finite Element Analysis (FEA) should be performed on the power transmission system to evaluate stresses induced.

2. Need for This Problem Statement

Real-World Application:

Sieving machines are widely used in agriculture, construction, pharmaceuticals, and food industries for separating materials like sand, grains, powders, and cement.

Increased Efficiency:

Manual sieving is time-consuming and inconsistent. A mechanical sieving machine improves both speed and accuracy.

Technical Learning:

The project integrates concepts from mechanical vibrations, power transmission, structural analysis, and machine design.

Cost-Effective Solution:

Most industrial vibrating sievers are expensive. Designing a compact, low-cost system helps small businesses and labs.

User Safety & Convenience:

A stable and easy-to-operate machine reduces operator fatigue and ensures consistent product quality.

3. Definition of the Problem

- Capacity Requirement:
The machine must process 2–3 kg of mixed material per batch.
- Separation Efficiency:
It should effectively separate into at least two size fractions, based on mesh size.
- Vibration Mechanism:
A motor-driven eccentric cam or crank-slider must produce uniform vibration.
- Structural Stability:
The supporting frame must withstand continuous vibration without permanent deformation.
- User-Friendly Operation:
It should allow easy mesh replacement, material loading, and collection.

4. Synthesis

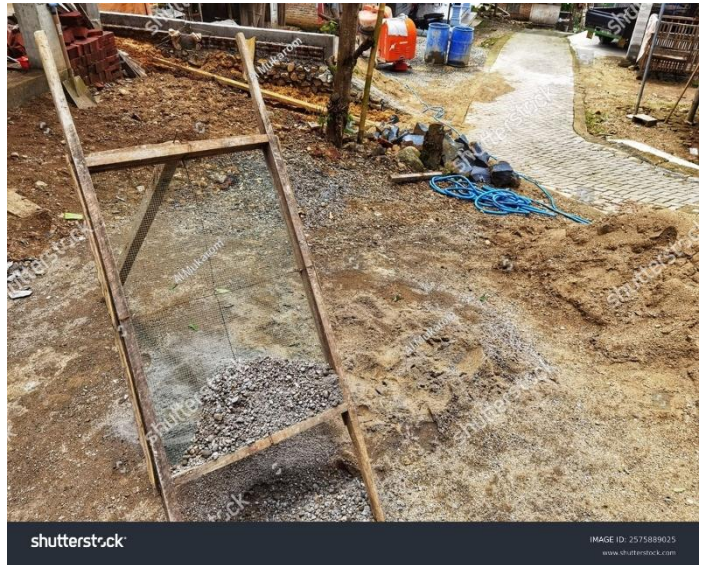
4.1 Initial Idea

The first idea was to use a reciprocating hand-operated system similar to traditional manual sieves.

However, several issues were identified:

- Inconsistent vibration amplitude due to manual operation.
- Low output, not suitable for bulk separation.
- Operator fatigue and varying efficiency.
- Lack of structural stability for heavy or continuous usage.

Due to these limitations, the idea was rejected.



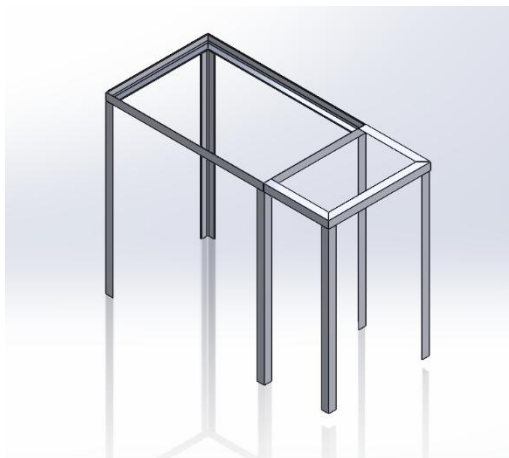
5. Final Solution

The final design uses a motor-driven crank–slider mechanism to generate controlled reciprocating motion for the sieve tray. The crank–slider provides a simple way to convert rotary motion of the motor into a linear reciprocating motion.

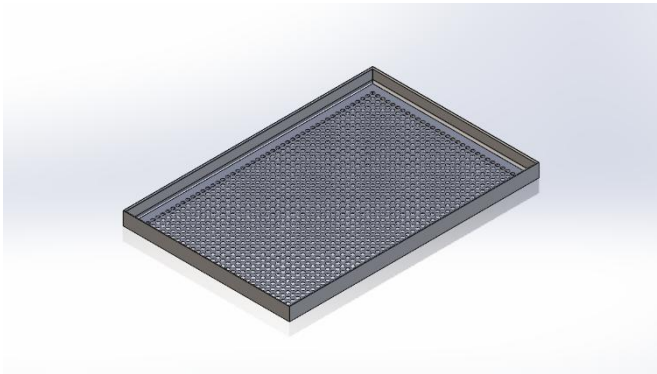
Basic Structure

The system consists of:

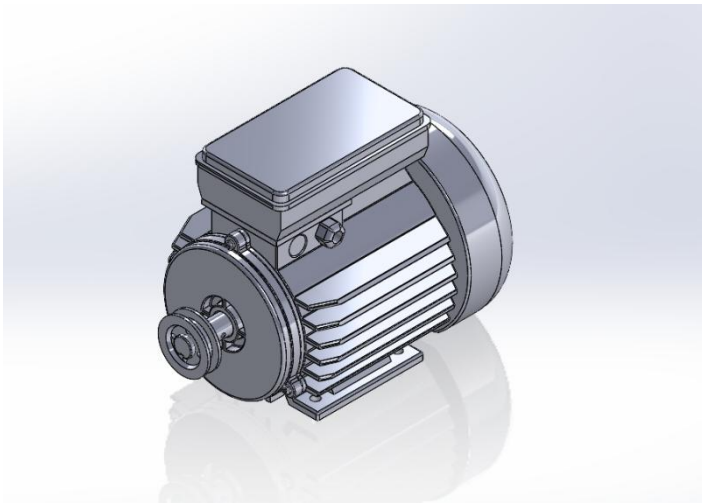
- Rectangular mild steel frame



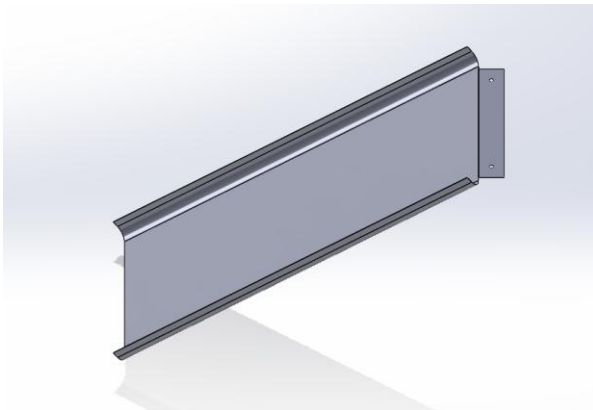
- A sieve box with replaceable mesh plates



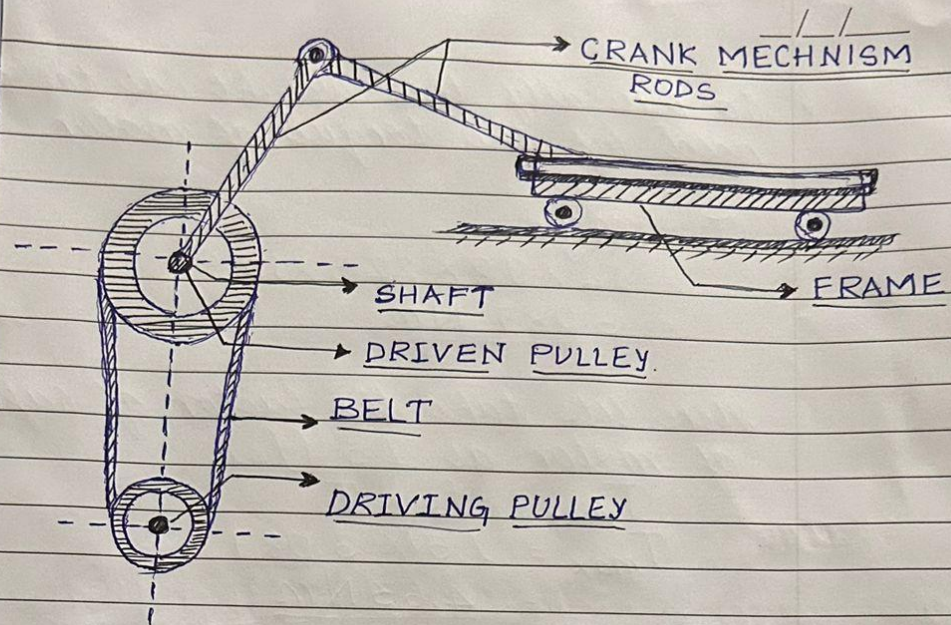
- A crank–slider assembly driven by a 1 HP electric motor



- Collecting tray below the mesh



6. Design Calculations:



- mass of driving pulley: 0.158 Kg
- mass of driven pulley: 3.757 Kg
- mass of the belt: 0.523 Kg
- mass of the shaft: 1.046 Kg
- mass of crank rods: 0.381 Kg
- mass of frame: 5.301 Kg
- mass of material sieved: 3. Kg

→ Total mass of everything above: 14.009 Kg
(m)

The radius of driving pulley: 22.5 mm
(r)

→ So basically this weight has to be balanced by the torque of motor.

$$\begin{aligned} T &= mgr \\ &= 14 \times 9.8 \times 22.5 \times 10^{-3} \\ &= \underline{3.087 \text{ Nm}} \end{aligned}$$

Here let's take the factor of safety of motor as 1.5 (f_s)

$$\begin{aligned} T_{\max} &= f_s \times T \\ T_{\max} &= 1.5 \times 3.087 \\ &= \underline{4.63 \text{ Nm}} \end{aligned}$$

For this value of Torque max we get the rpm from standard motor chart. From chart we get rpm as 1500 rpm

$$\begin{aligned} \text{Power} \leftarrow P &= T \times \omega \\ &= 4.63 \times 2\pi \times \frac{1500}{60} \\ &= 727.27 \text{ W} \\ &(\text{or}) \approx \underline{1 \text{ Horse power}} \end{aligned}$$

As power is near to 750 W which is correspondent to 1 HP, so hence we choose 1 HP motor 1500 rpm

Also we can calculate the belt length

$$\rightarrow D \text{ (The diameter of driven pulley)} = 0.145 \text{ m}$$

$$\rightarrow d \text{ (The diameter of driving pulley)} = 0.045 \text{ m}$$

$$\rightarrow \text{speed ratio} = \frac{d}{D} = \frac{0.045}{0.145} = 0.31$$

The angular velocity with which
big (or) driven shaft runs is
 $\rightarrow N(\text{rpm})$ of driving pulley

$$= \frac{1500 \times 0.31 \times 2 \times \pi}{60}$$

$$= 48.69 \text{ rad/s}$$

$\rightarrow$ The distance between driving and
driven pulley centres are

$$x = 0.47 \text{ m}$$

Length of Belt

$$= \frac{\pi}{2} (D + d) + 2(x) + \frac{(D - d)^2}{4x}$$

$$= \frac{\pi}{2} (0.145 + 45 \times 10^{-3}) + 2(0.47) + \frac{(0.145 - 0.045)^2}{4 \times 0.47}$$

$$= 1.25 \text{ m}$$